Topic 5 – Separate chemistry 1

Transition metals, alloys and corrosion

Students should:		Maths skills
5.1C	Recall that most metals are transition metals and that their typical properties include: a high melting point b high density c the formation of coloured compounds d catalytic activity of the metals and their compounds as exemplified by iron	
5.2C	Recall that the oxidation of metals results in corrosion	
5.3C	Explain how rusting of iron can be prevented by: a exclusion of oxygen b exclusion of water c sacrificial protection	
5.4C	Explain how electroplating can be used to improve the appearance and/or the resistance to corrosion of metal objects	
5.5C	Explain, using models, why converting pure metals into alloys often increases the strength of the product	5b
5.6C	Explain why iron is alloyed with other metals to produce alloy steels	
5.7C	Explain how the uses of metals are related to their properties (and vice versa), including aluminium, copper and gold and their alloys including magnalium and brass	

Suggested practicals

- Carry out an activity to show that transition metal salts have a variety of colours.
- Investigate the rusting of iron.
- Electroplate a metal object.
- Make an alloy or investigate the properties of alloys.

Quantitative analysis

Students should:		Maths skills
5.8C	**Calculate the concentration of solutions in mol dm^{-3} and convert concentration in g dm^{-3} into mol dm^{-3} and vice versa**	1a, 1b, 1c, 1d 2a 3b, 3c
5.9C	*Core Practical: Carry out an accurate acid-alkali titration, using burette, pipette and a suitable indicator*	
5.10C	**Carry out simple calculations using the results of titrations to calculate an unknown concentration of a solution or an unknown volume of solution required**	1a, 1c, 1d 2a, 2b 3a, 3b, 3c
5.11C	Calculate the percentage yield of a reaction from the actual yield and the theoretical yield	1a, 1c, 1d 2a 3b, 3c
5.12C	Describe that the actual yield of a reaction is usually less than the theoretical yield and that the causes of this include: a incomplete reactions b practical losses during the experiment c competing, unwanted reactions (side reactions)	
5.13C	Recall the atom economy of a reaction forming a desired product	
5.14C	Calculate the atom economy of a reaction forming a desired product	1a, 1c, 1d 2a 3c
5.15C	**Explain why a particular reaction pathway is chosen to produce a specified product, given appropriate data such as atom economy, yield, rate, equilibrium position and usefulness of by-products**	
5.16C	**Describe the molar volume, of any gas at room temperature and pressure, as the volume occupied by one mole of molecules of any gas at room temperature and pressure** **(The molar volume will be provided as 24 dm^3 or 24000 cm^3 in calculations where it is required)**	
5.17C	**Use the molar volume and balanced equations in calculations involving the masses of solids and volumes of gases**	1a, 1c, 2a 3b, 3c
5.18C	**Use Avogadro's law to calculate volumes of gases involved in a gaseous reaction, given the relevant equation**	1a, 1c, 1d

Use of mathematics

- Arithmetic computation when calculating yields and atom economy (1a and 1c).
- Arithmetic computation, ratio, percentage and multistep calculations permeates quantitative chemistry (1a, 1c and 1d).
- Change the subject of a mathematical equation (3b and 3c).
- Provide answers to an appropriate number of significant figures (2a).
- **Convert units where appropriate particularly from mass to moles (1c)**.

Suggested practicals

- Prepare a substance and calculate the percentage yield, given the theoretical yield.
- Determine the volume of one mole of hydrogen gas by using the reaction of magnesium with hydrochloric acid.

Dynamic equilibria

Students should:		Maths skills
5.19C	Describe the Haber process as a reversible reaction between nitrogen and hydrogen to form ammonia	
5.20C	**Predict how the rate of attainment of equilibrium is affected by:** **a changes in temperature** **b changes in pressure** **c changes in concentration** **d use of a catalyst**	
5.21C	**Explain how, in industrial reactions, including the Haber process, conditions used are related to:** **a the availability and cost of raw materials and energy supplies** **b the control of temperature, pressure and catalyst used produce an acceptable yield in an acceptable time**	
5.22C	Recall that fertilisers may contain nitrogen, phosphorus and potassium compounds to promote plant growth	
5.23C	Describe how ammonia reacts with nitric acid to produce a salt that is used as a fertiliser	
5.24C	Describe and compare: a the laboratory preparation of ammonium sulfate from ammonia solution and dilute sulfuric acid on a small scale b the industrial production of ammonium sulfate, used as a fertiliser, in which several stages are required to produce ammonia and sulfuric acid from their raw materials and the production is carried out on a much larger scale (details of the industrial production of sulfuric acid are not required)	

Suggested practicals

- Prepare a sample of ammonium sulfate from ammonia solution and dilute sulfuric acid.

Chemical cells and fuel cells

Students should:		Maths skills
5.25C	Recall that a chemical cell produces a voltage until one of the reactants is used up	
5.26C	Recall that in a hydrogen–oxygen fuel cell hydrogen and oxygen are used to produce a voltage and water is the only product	
5.27C	Evaluate the strengths and weaknesses of fuel cells for given uses	